Project 1 - Tribble Population Modeling

APPM 2360

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Introduction: Questions 1-2: Model and Interpretation

The term ay is a linear growth term representing tribble reproduction. The term −by^2 is nonlinear and is a population-dependent loss term that represents the limitations for tribbles (ex: competition for resources). The term −H(y)=(−py^3)/[(y^3)+q] is a nonlinear growth/decay term representing tribbles removed by Klingon hunting. The hunting rate increases with tribble population but approaches a maximum rate of p for large populations.

When y gets very small H(y), the hunting rate will also be very small as the animals to hunt become fewer so does the rate the Klingons need to hunt. As y gets very big and approaches infinity, the hunting rate will increase until the population is very large and reaches a maximum as the population won't increase indefinitely as the rate will grow with it.

Question 3: Parameter Units

$$dy/dt = \text{population} / \text{year}$$

$$ay = a(\text{population}) = \text{population} / \text{year}$$

$$a = 1 / \text{year} = (\text{year})^{(-1)}$$

$$by^2 = b(\text{population})^2 = \text{population} / \text{year}$$

$$b = 1 / [(\text{population})(\text{year})] = (\text{population})^{(-1)}(\text{year})^{(-1)}$$

Figure 1. Work determining the units of the parameters a and b.

Question 4: Analytical Solution and Equilibria

Solution Without Hunting

y(t)=(ay[naught](e^at))/(a+by[naught]((e^at)-1))

a/b represents the long-term population of tribbles when there is no hunting, measured in hundreds of tribbles

Solution With Hunting

integral [1/(ay - (by^2) - (py/(y^3 + q))]dy

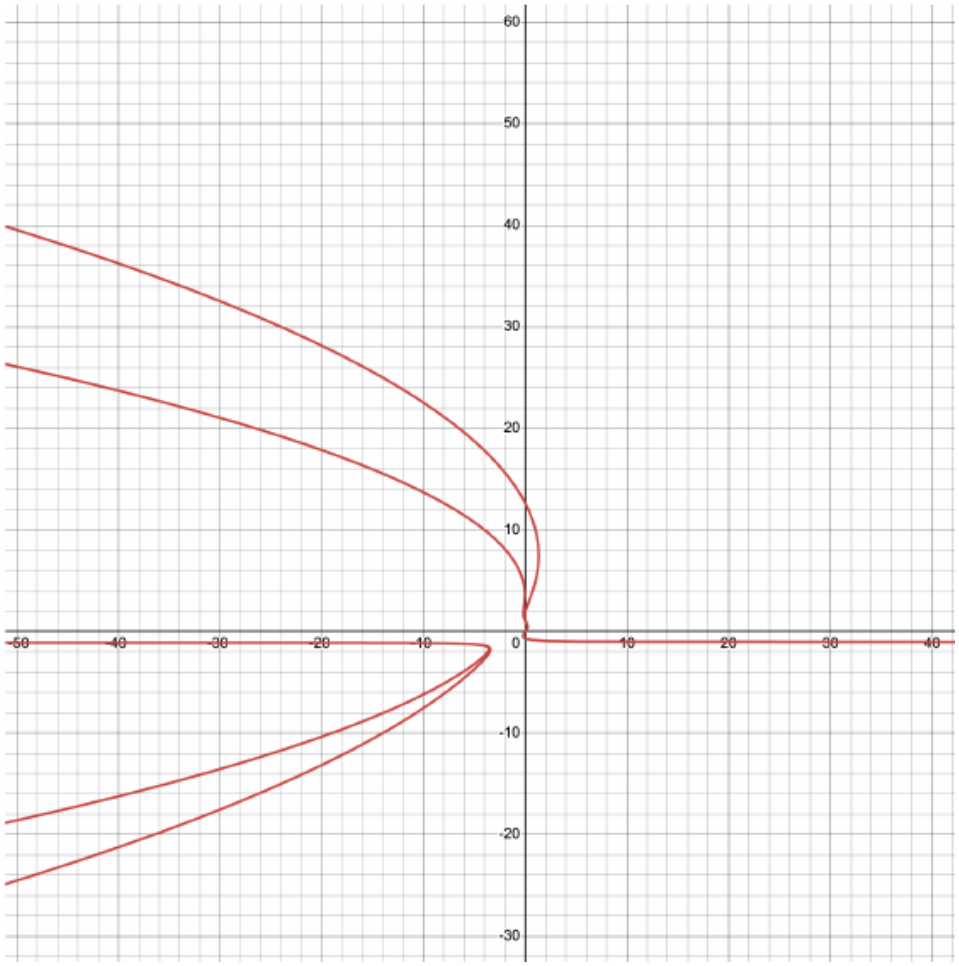
Yes, the equation can still be solved by separation of variables. Using partial fraction decomposition to transform this function into an integrable form and factoring the denominator makes it solvable.

Question 5: Equilibrium Solutions

The equilibrium of the ODE y'=f(y) is when y(t) = y* for f(y*) = 0 is an equilibrium solution.

Setting y'=f(y) to y'=0 so that when f(y)=0 is when an equilibrium point occurs.

Questions 6-7: Function Plots and Direction Fields



Q6: b = 0.005, b = 0.05 & b = 0.10. Plot f(y) & find equilibrium solutions (enter b as parameter/slider).

$$H(y) = py^3 / (y^3 + q),\ \ p = 1.5,\ \ q = 1.25,\ \ a = 0.75.$$

$$f(y) = ay - by^2 - H(y).$$

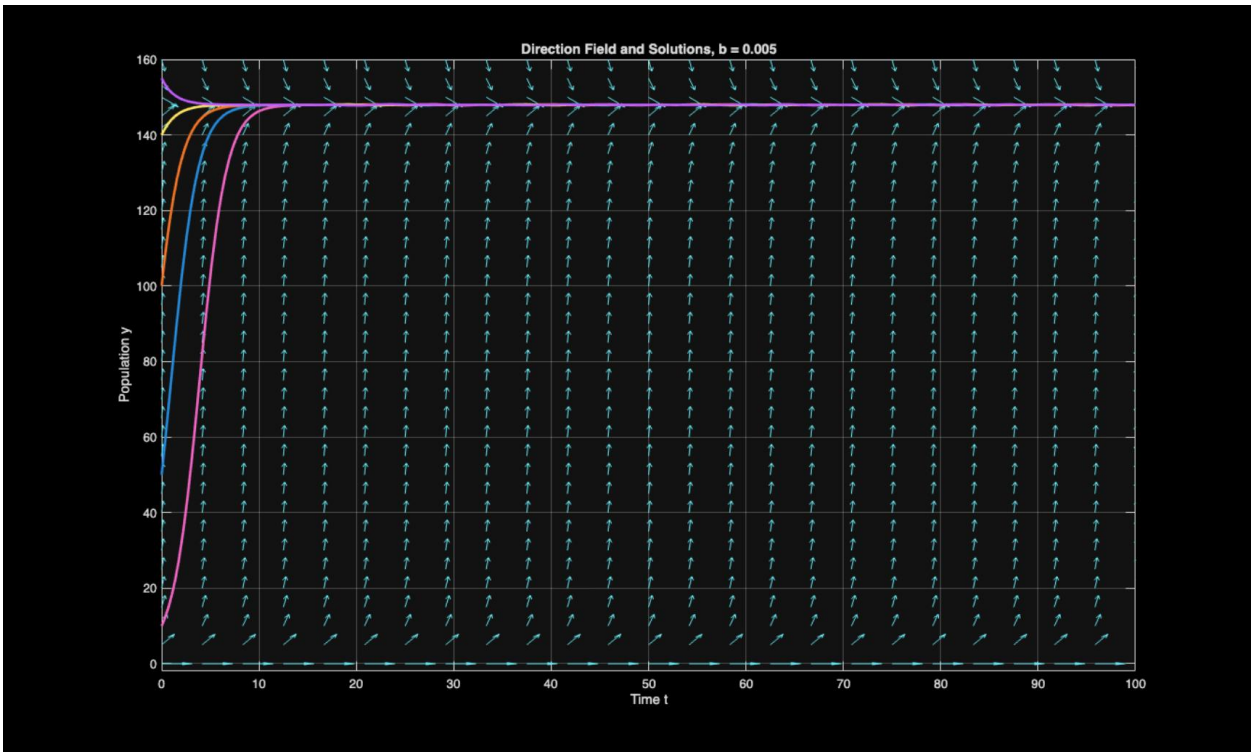
For b = 0.005: $f(y) = 0.75y - 0.005y^2 - (1.5y^3)/(y^3 + 1.25) \Rightarrow$ E.S.: $f(0) = 0$.

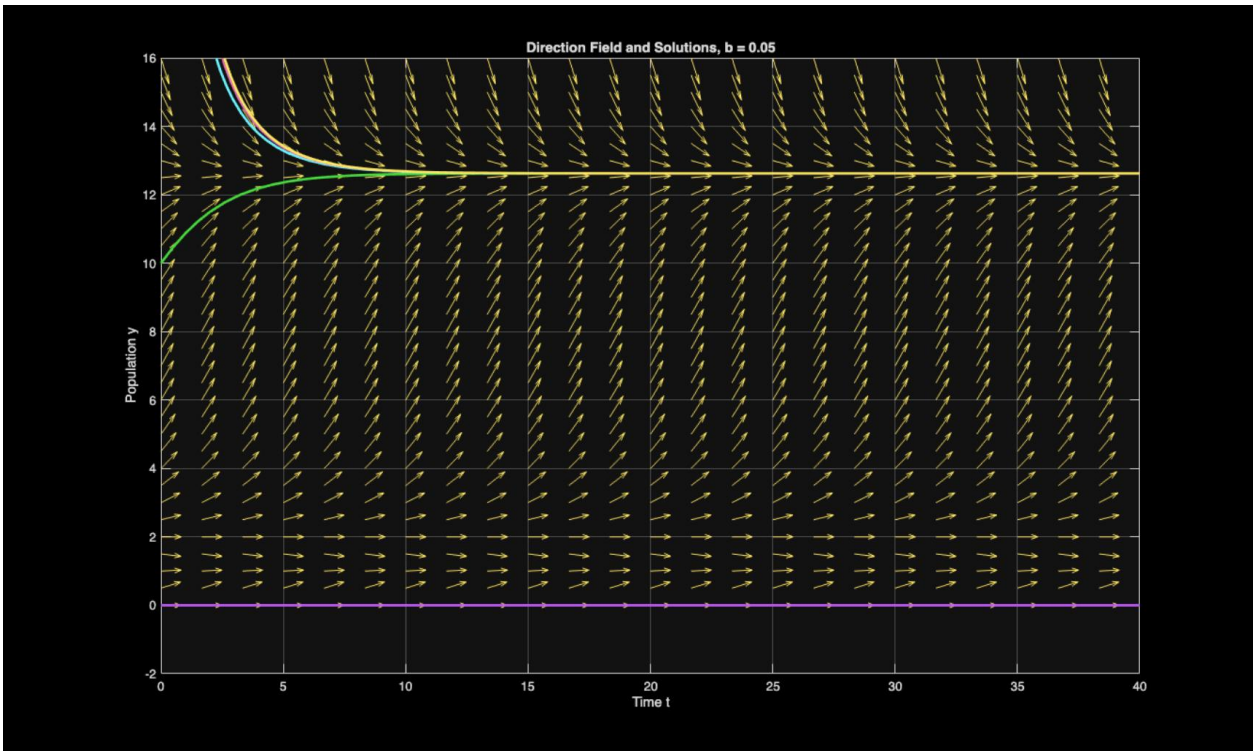
For b = 0.05: $f(y) = 0.75y - 0.05y^2 - (1.5y^3)/(y^3 + 1.25) \Rightarrow$ E.S.: $f(0) = 0$.

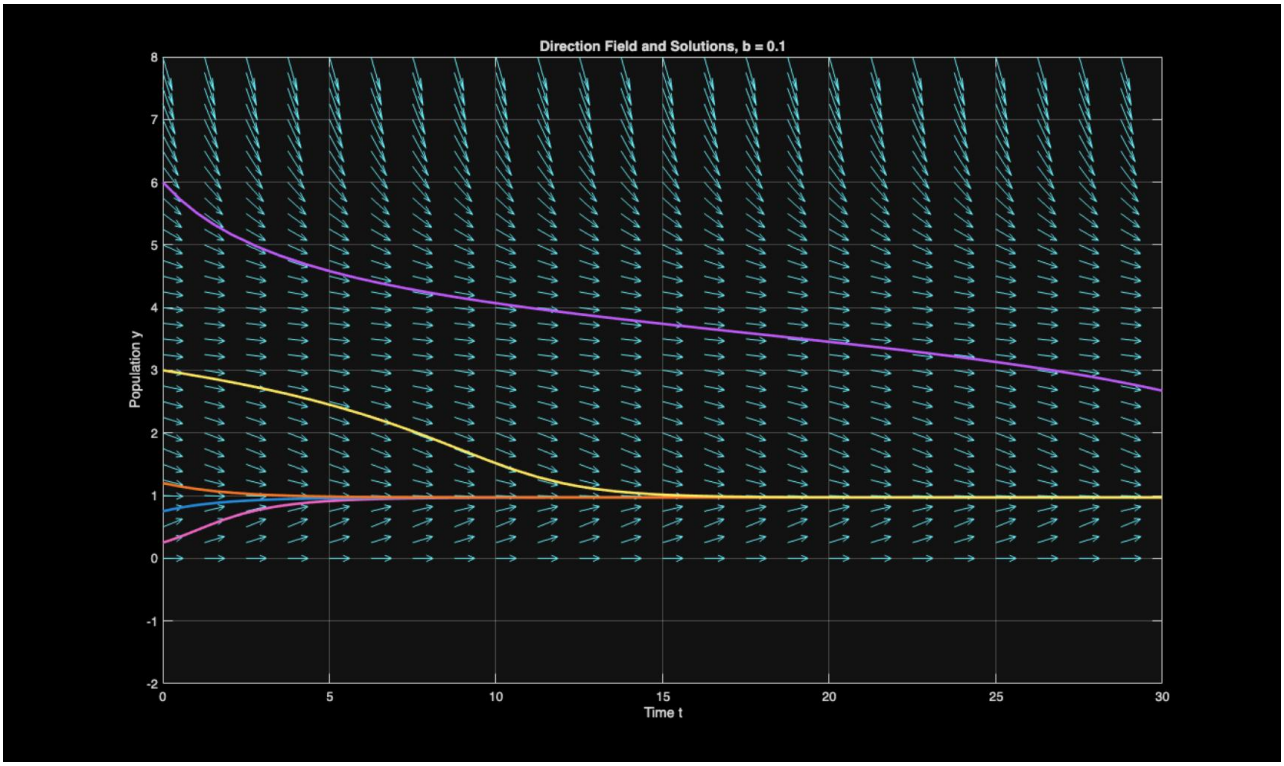
For b = 0.10: $f(y) = 0.75y - 0.10y^2 - (1.5y^3)/(y^3 + 1.25) \Rightarrow$ E.S.: $f(0) = 0$.

Figure 2. Function plots used to identify equilibrium solutions for the selected b values.

Figure 3. Direction fields and solution curves for the selected b values.


Direction Field and Solutions, b = 0.005
Population y
Time t
0
20
40
60
80
100
120
140
160
0
10
20
30
40
50
60
70
80
90
100


Direction Field and Solutions, b = 0.05
Population y
Time t
-2
0
2
4
6
8
10
12
14
16
0
5
10
15
20
25
30
35
40



Question 8: Long-Term Behavior and Stability

Equilibrium Solution b	Y	Stability
0.005	0	Unstable
0.005	147.977	Stable
0.05	0	Unstable
0.05	12.6254	Stable
0.05	1.07758	Stable
0.05	1.9999	Unstable
0.10	0	Unstable
0.10	0.971	Stable

b	Basin of Attraction	Description
0.005	(0,infinity)	Every positive initial population goes towards it
0.05	Y0 = 1.991	Only at the unstable
0.10	(0, infinity)	Every positive initial population goes towards it

The solution curves and direction fields represented on the graphs are used to determine and analyze long-term behavior of the tribble populations.

For graph b=0.005, the positive initial populations all approach the same stable equilibrium solution y=147.973. If the population starts above the equilibrium solution, it decreases. If the population starts below the equilibrium solution, then it will increase.

For graph b=0.05, if the population starts below the y=1.99 equilibrium, then it will approach the y=1.077 equilibrium. If the population starts above the y=1.99 equilibrium, it will then approach the y=12.6 equilibrium.

For graph b=0.10, if the initial population starts below the y=1 equilibrium, it will increase, and if it starts above it will decrease to the y=1 equilibrium solution.

Question 9: Stocking Recommendation

We'll choose to maximize sustainable cultivation to grow and harvest as many tribbles as possible without crashing the population into extinction. Since b is the death/loss rate, grey tribbles (b=0.10) die the fastest, making them the easiest to control. Brown tribbles (b=0.005) would multiply unrestrained. The best species to stock is brown tribbles; any positive number greater than the initial population Y0, will satisfy the priority of maximizing a sustainable population. A lower b parameter signifies high resilience or rapid net reproduction. For the brown tribbles (b=0.005), the table from question 8 shows an unstable equilibrium at Y=0 and a massive stable equilibrium at Y=147.977. Because its basin of attraction is the entire positive range (0, infinity), stocking even a minimal amount of brown tribbles guarantees the population will thrive and steadily grow to a large population, producing a carrying capacity of approximately 148 tribble units without collapsing.

White tribbles (b=0.05) aren't a good choice to stock because they are highly unpredictable due to having 2 equilibrium solutions. They possess an unstable threshold near Y=1.9999. Having this sensitivity means that if our initial stocking count accidentally drops below Y=1.9999, the population will crash down to the lower stable state instead of reaching the higher capacity.

Question 10: Seasonal Hunting Model

The solutions eventually settle into a repeating seasonal pattern. The main change to the model is that the hunting function now depends on both the population y and time t, making the differential equation nonautonomous. Since the t and y terms cannot be separated and separation of variables can no longer be used to find an analytical solution, numerical methods are needed instead (Euler's Method). The only equilibrium solution that remains is y=0. The previous nonzero equilibria are no longer constant because the amount of hunting changes throughout the year.

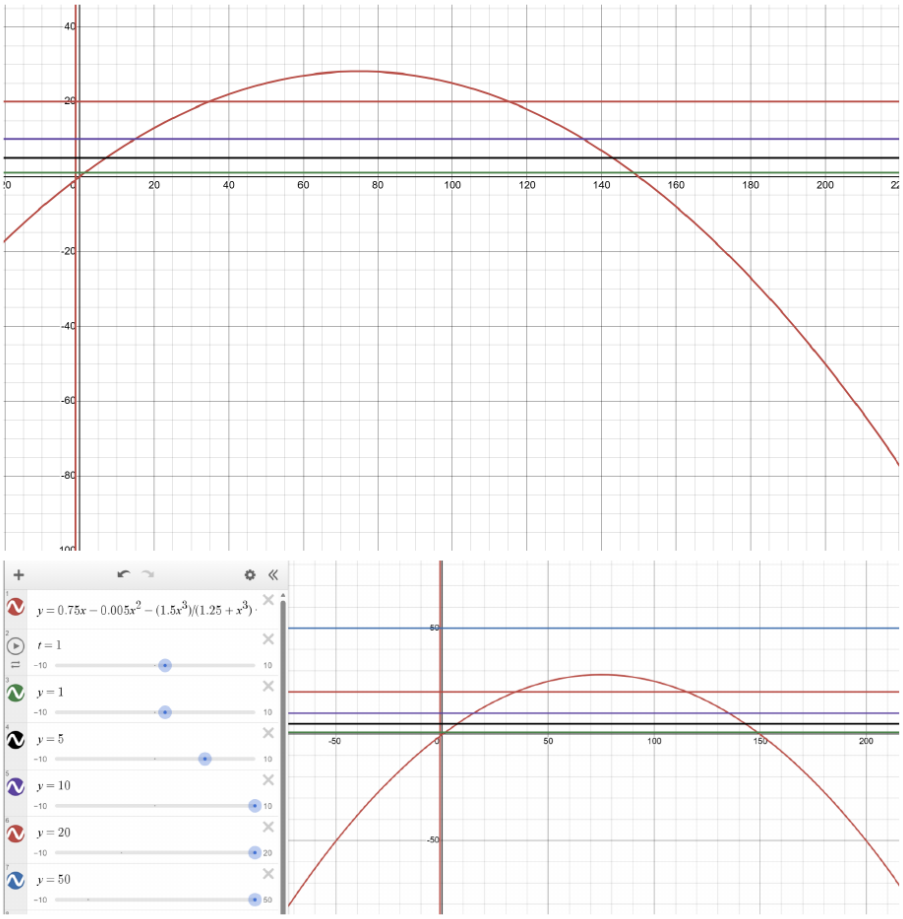
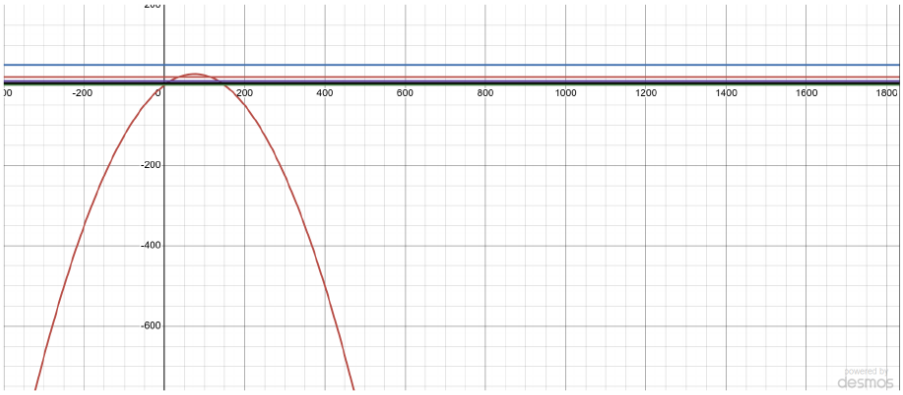


Figure 4. Seasonal hunting results for b=0.005.



b=0.05

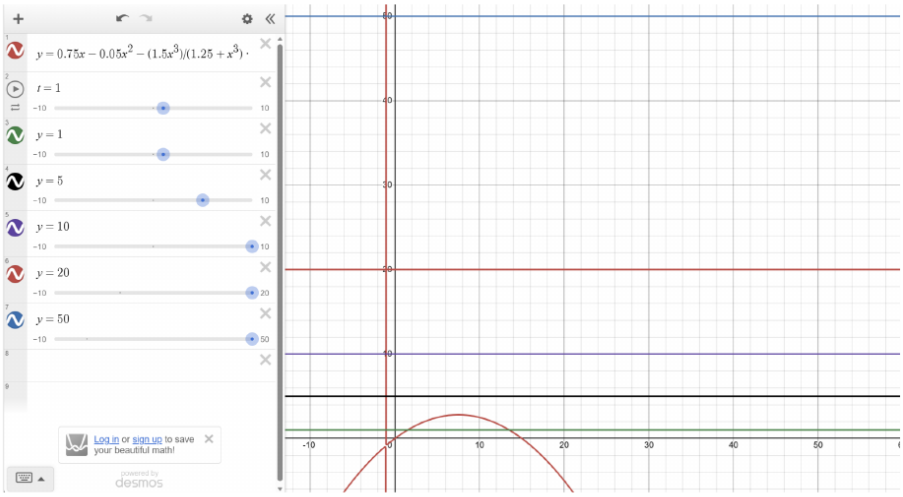


Figure 5. Seasonal hunting results for b=0.05.

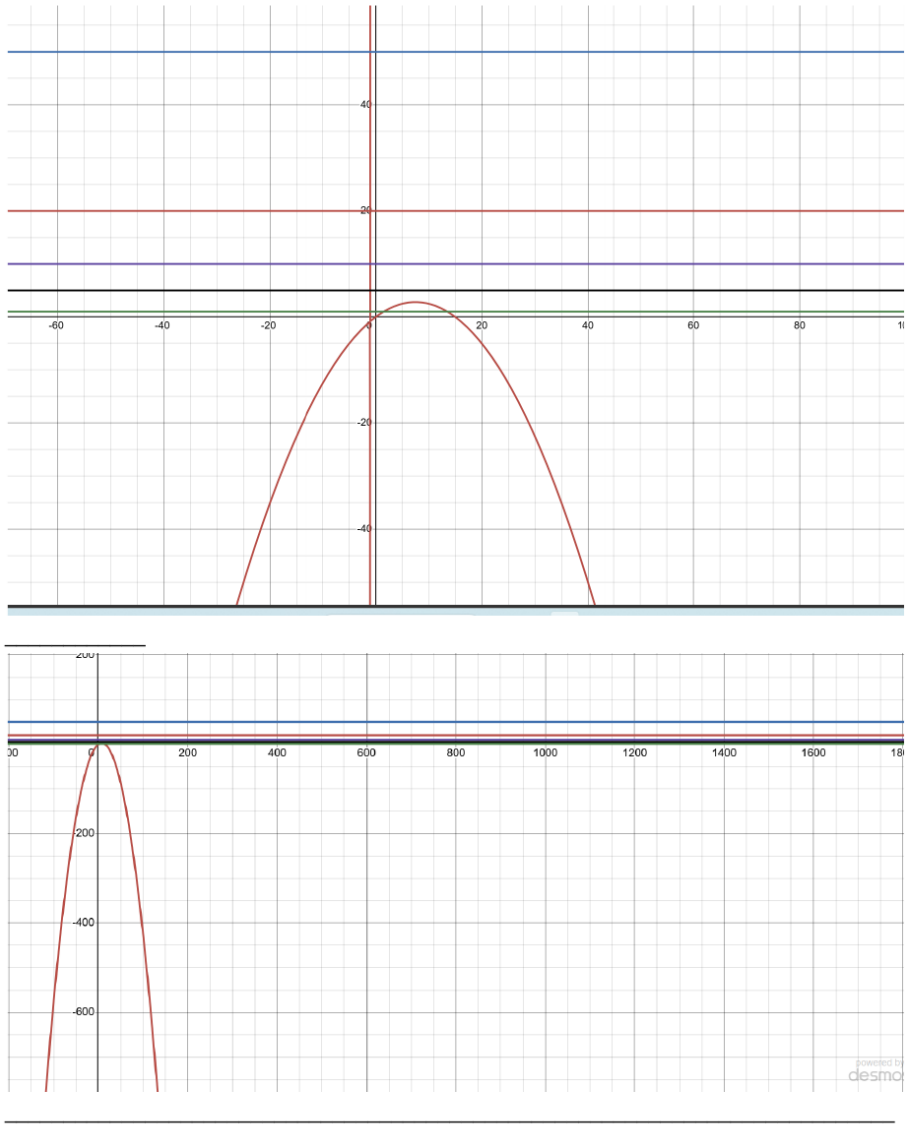


Figure 6. Seasonal hunting results for b=0.10.

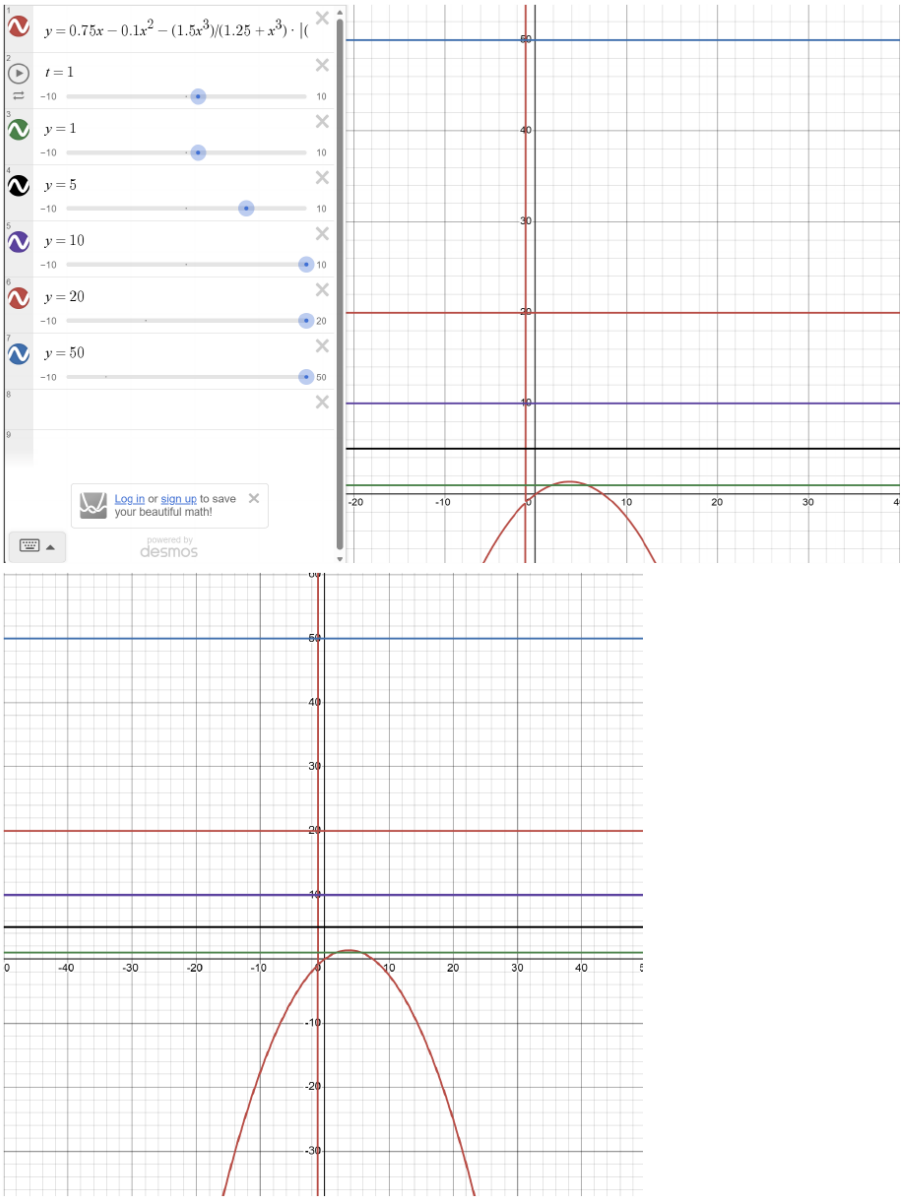


Figure 7. Additional seasonal hunting solution plots.

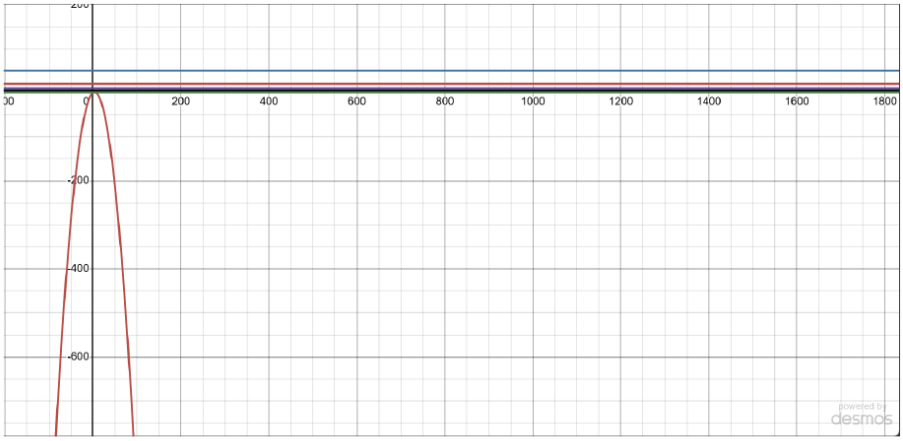
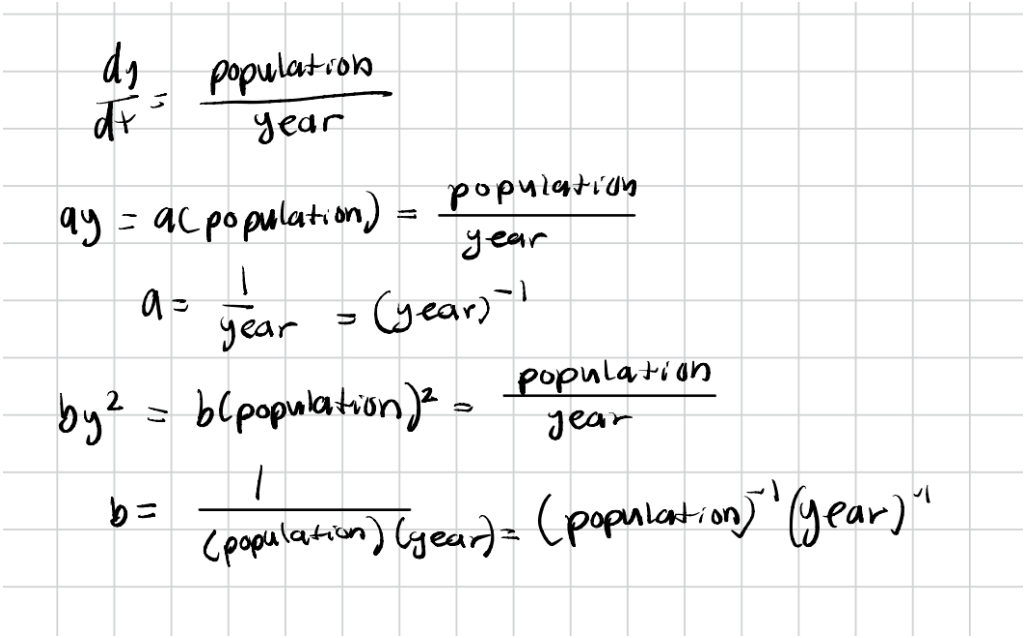


Figure 8. Long-time seasonal hunting plot.

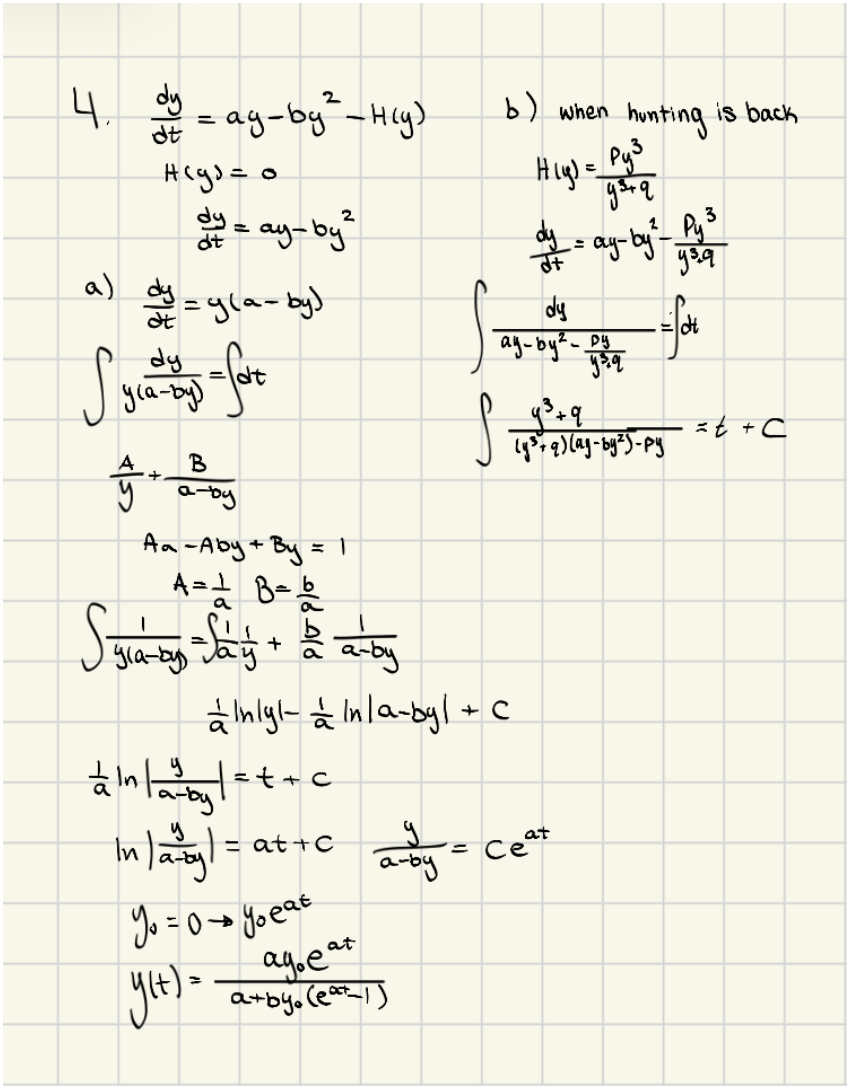
Question 11: Conclusion

A potential weakness we noticed is that the model is only regarding Tribbles and Klingons, so this doesn't consider potentially different predators or environmental factors, such as sickness or disease, that could also affect the population of the Tribbles. These are weaknesses because the model has no parameters changing, while the real world would have other parameters that'd affect the model. We would improve this problem by including these real-world parameters or other models that would represent them. However, because of the situation and problem we are trying to solve, our model is an efficient representation.

Appendix A: Calculations



Appendix Figure A1. Parameter-unit calculations.



Appendix Figure A2. Separation-of-variables calculations for the model.

Appendix Figure 3. Function Plots and Direction Fields

Q2: $H(y) = \frac{1.5\,y^3}{y^3+1.25}$ as $\lim_{y\to 0} H(y) = 0$

$$\lim_{y\to\infty} \frac{1.5y^3}{y^3+1.25} = 1.5$$

Q5: a) $y' = f(y) \rightarrow y(t) = y^*$ for $f(y^*) = 0$ equilibrium solution

b) if $y' = 0$ then $f(y) = 0$ is an equilibrium point

Q6: $b = 0.005$, $= 0.05$ & $= 0.10$ plot $f(y)$ & find equilibrium solutions (enter b as parameter/slider)

$$f(y) = ay - by^2 - H(y) \qquad H_y = \frac{py^3}{y^3+q} \qquad p = 1.5,\ q = 1.25,\ a = 0.75$$

$f(y) = 0.75y - 0.005y^2 - \frac{1.5y^3}{y^3+1.25}$ $\Rightarrow$ E.S.: $f(0) = 0$

$f(y) = 0.75y - 0.05y^2 - \frac{1.5y^3}{y^3+1.25}$ $\Rightarrow$ E.S.: $f(0) = 0$

$f(y) = 0.75y - 0.10y^2 - \frac{1.5y^3}{y^3+1.25}$ $\Rightarrow$ E.S.: $f(0) = 0$

Appendix B: Code

```
%p, q, a constants
a = 0.75;
p = 1.5;
q = 1.25;
%bvalues
bvalues = [0.005 0.05 0.10];
for k = 1:length(bvalues)
        b = bvalues(k);
%DE
f = @(t,y) a*y - b*y.^2 - (p*y.^3)./(y.^3 + q);
%graph size constraints
if b == 0.005
        tmax = 100;
        ymax = 160;
        ystep = 5;
        y0values = [10 50 100 140 155];
elseif b == 0.05
        tmax = 40;
        ymax = 16;
        ystep = 0.5;
% ICs
y0values = [0 10 50 100 140 147.973 155];
else
        tmax = 30;
        ymax = 8;
        ystep = 0.25;
%%%%%why these numbers
y0values = [0.25 0.75 1.2 3 6];
end
%make the grid
tvals = linspace(0,tmax,25);
yvals = 0:ystep:ymax;
[T,Y] = meshgrid(tvals,yvals);
%slope
S = f(T,Y);
%arrows
U = ones(size(S));
L = sqrt(U.^2 + S.^2);
U = U./L;
V = S./L;
%make the figure
figure(k)
%direction feilds
quiver(T,Y,U,V,0.5)
hold on
%plot the solutions
for y0 = y0values
[t,y] = ode45(f,[0 tmax],y0);
plot(t,y,'LineWidth',2)
end
%labels
```

```
xlabel('Time t')
ylabel('Population y')
%title
title(['Direction Field and Solutions, b = ', num2str(b)])
%limits
xlim([0 tmax])
ylim([-2 ymax])
grid on
end
```